Week1 : Objectives

Upon completion of this unit, students will be able to understand and explain the following sub topics :

- Introduction to Bags,
- Array-based Bag Implementation,
- Linked Data Bag Implementation

What Is an Iterator?

- An object that traverses a collection of data
- During iteration, each data item is considered once
  - Possible to modify item as accessed
- Should implement as a distinct class that interacts with the ADT

The ADT Bag

- Definition
  - A finite collection of objects in no particular order
  - Can contain duplicate items
- Possible behaviors
  - Get number of items
  - Check for empty
  - Add, remove objects

CRC Card

- Bag for CRC Card

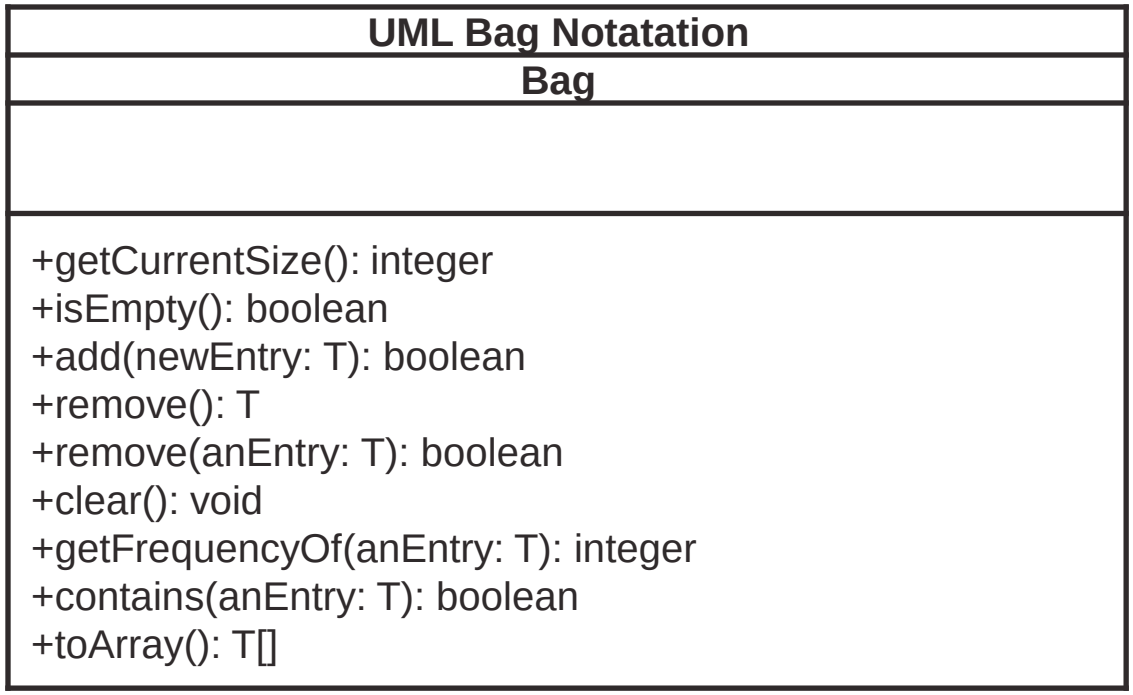
CRC Card
Bag
Responsibilities
Get the number of items currently in the bag
See whether the bag is empty
Add a given object to the bag
Remove an unspecified object from the bag
Remove a particular object from the bag, if possible
Remove all objects from the bag
Count the number of times a certain object occurs in the bag
Test whether the bag contains a particular object
Look at all objects that are in the bag
Collaborations
The class of objects that the bag can contain

Specifying a Bag

- Describe its data and specify in detail the methods
- Options that we can take when add cannot complete its task:
  - Do nothing
  - Leave bag unchanged, but signal client
- Note which methods change the object or do not

Using UML Notation to Specify a Class

- The following is Bag UML diagram



Design Decision

- What to do for unusual conditions?
- Assume it won’t happen
- Ignore invalid situations
- Guess at the client’s intention
- Return value that signals a problem
- Return a boolean
- Throw an exception

An Interface (Part 1)

- **LISTING 1-1 A Java interface for a class of bags**

```
/** An interface that describes the operations of a bag of objects. */
public interface BagInterface<T>
{
      /** Gets the current number of entries in this bag.
       @return  The integer number of entries currently in the bag. */
      public int getCurrentSize();

      /** Sees whether this bag is empty.
       @return  True if the bag is empty, or false if not. */
      public boolean isEmpty();

      /** Adds a new entry to this bag.
         @param newEntry  The object to be added as a new entry.
         @return  True if the addition is successful, or false if not. */
      public boolean add(T newEntry);

      /** Removes one unspecified entry from this bag, if possible.
    @return  Either the removed entry, if the removal.
          was successful, or null. */
      public T remove();
```

An Interface (Part 2)

- This is the bag java interface

```
    /** Removes one occurrence of a given entry from this bag, if possible.
   @param anEntry  The entry to be removed.
   @return  True if the removal was successful, or false if not. */
  public boolean remove(T anEntry);

    /** Removes all entries from this bag. */
    public void clear();

    /** Counts the number of times a given entry appears in this bag.
     @param anEntry  The entry to be counted.
     @return  The number of times anEntry appears in the bag. */
    public int getFrequencyOf(T anEntry);

    /** Tests whether this bag contains a given entry.
     @param anEntry  The entry to find.
     @return  True if the bag contains anEntry, or false if not. */
    public boolean contains(T anEntry);

    /** Retrieves all entries that are in this bag.
     @return  A newly allocated array of all the entries in the bag.
         Note: If the bag is empty, the returned array is empty. */
    public T[] toArray();
} // end BagInterface
```

Using the ADT Bag

```
/** A class that maintains a shopping cart for an online store. */
public class OnlineShopper
{
        public static void main(String[] args)
        {
    Item[] items = {new Item("Bird feeder", 2050),
                    new Item("Squirrel guard", 1547),
                    new Item("Bird bath", 4499),
                    new Item("Sunflower seeds", 1295)};

    BagInterface<Item> shoppingCart = new Bag<>();
    int totalCost = 0;
   // Statements that add selected items to the shopping cart:
    for (int index = 0; index < items.length; index++)
    {
      Item nextItem = items[index]; // Simulate getting item from shopper
      shoppingCart.add(nextItem);
      totalCost = totalCost + nextItem.getPrice();
    } // end for

    // Simulate checkout
    while (!shoppingCart.isEmpty())
            System.out.println(shoppingCart.remove());

        System.out.println("Total cost: " + "\t$" + totalCost / 100 + "." +
                    totalCost % 100);
        } // end main
} // end OnlineShopper
```

Program Output

```
Sunflower seeds $12.95
Bird bath       $44.99
Squirrel guard      $15.47
Bird feeder        $20.50
Total cost:     $93.91
```

Example: A Piggy Bank

```
/** A class that implements a piggy bank by using a bag. */
public class PiggyBank
{
    private BagInterface<Coin> coins;

    public PiggyBank()
    {
    coins = new ArrayBag<>();
    } // end default constructor

    public boolean add(Coin aCoin)
    {
    return coins.add(aCoin);
    } // end add

    public Coin remove()
    {
    return coins.remove();
    } // end remove

    public boolean isEmpty()
    {
    return coins.isEmpty();
    } // end isEmpty
```

Example: Using A Piggy Bank (Part 1)

```
/** A class that demonstrates the class PiggyBank. */
public class PiggyBankExample
{
    public static void main(String[] args)
    {
    PiggyBank myBank = new PiggyBank();

    addCoin(new Coin(1, 2010), myBank);
    addCoin(new Coin(5, 2011), myBank);
    addCoin(new Coin(10, 2000), myBank);
    addCoin(new Coin(25, 2012), myBank);

    System.out.println("Removing all the coins:");
    int amountRemoved = 0;

    while (!myBank.isEmpty())
    {
      Coin removedCoin = myBank.remove();
      System.out.println("Removed a " + removedCoin.getCoinName() + ".");
      amountRemoved = amountRemoved + removedCoin.getValue();
    } // end while
```

Example: Using A Piggy Bank (Part 2)

- addCoin method

```
private static void addCoin(Coin aCoin, PiggyBank aBank)
{
if (aBank.add(aCoin))
    System.out.println("Added a " + aCoin.getCoinName() + ".");
else
    System.out.println("Tried to add a " + aCoin.getCoinName() +
                       ", but couldn't");
} // end addCoin
} // end PiggyBankEx
```

Program Output

```
Added a PENNY.
Added a NICKEL.
Added a DIME.
Added a QUARTER.
Removing all the
coins:
Removed a QUARTER.
Removed a DIME.
Removed a NICKEL.
Removed a PENNY.
All done. Removed 41
cents.
```

Observations about Vending Machines

- Can perform only tasks machine's interface presents.
- You must understand these tasks
- Cannot access the inside of the machine
- You can use the machine even though you do not know what happens inside.
- Usable even with new insides.

Observations about ADT Bag

- Can perform only tasks specific to ADT
- Must adhere to the specifications of the operations of ADT
- Cannot access data inside ADT without ADT operations
- Use the ADT, even if don’t know how data is stored
- Usable even with new implementation.

Java Class Library: The Interface `Set`

- **A Java interface for a class of sets**

```
/** An interface that describes the operations of a set of objects.  */
public interface SetInterface<T>
{
        public int getCurrentSize();
        public boolean isEmpty();

        /** Adds a new entry to this set, avoiding duplicates.
           @param newEntry  The object to be added as a new entry.
           @return  True if the addition is successful, or
            false if the item already is in the set. */
        public boolean add(T newEntry);

        /** Removes a specific entry from this set, if possible.
      @param anEntry  The entry to be removed.
      @return  True if the removal was successful, or false if not. */
        public boolean remove(T anEntry);

        public T remove();
        public void clear();
        public boolean contains(T anEntry);
        public T[] toArray();
} // end SetInterface
```

BagInterface.java - E:\BlackUSB\GavilanCollege\Spring\CSIS27\WeeklyExamples\srcSol\Bag - Geany

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Symbols Documents

- Interfaces
  - BagInterface [7]
    - T [48]
    - add [20]
    - clear [33]
    - contains [43]
    - getCurrentSize [11]
    - getFrequencyOf [38]
    - isEmpty [15]
    - remove [25]
    - remove [30]
    - toArray [48]

LinkedBag1.java | LinkedBagDemo1.java | BagInterface.java

```java
/**
   An interface that describes the operations of a bag of objects.
   @author Frank M. Carrano
   @author Timothy M. Henry
   @version 5.0
*/
public interface BagInterface<T>
{
   /** Gets the current number of entries in this bag.
        @return  The integer number of entries currently in the bag. */
   public int getCurrentSize();

   /** Sees whether this bag is empty.
        @return  True if the bag is empty, or false if not. */
   public boolean isEmpty();

   /** Adds a new entry to this bag.
       @param newEntry  The object to be added as a new entry.
       @return  True if the addition is successful, or false if not. */
   public boolean add(T newEntry);

   /** Removes one unspecified entry from this bag, if possible.
      @return  Either the removed entry, if the removal
               was successful, or null. */
   public T remove();

   /** Removes one occurrence of a given entry from this bag, if possible.
      @param anEntry  The entry to be removed.
      @return  True if the removal was successful, or false if not. */
   public boolean remove(T anEntry);

   /** Removes all entries from this bag. */
   public void clear();

   /** Counts the number of times a given entry appears in this bag.
        @param anEntry  The entry to be counted.
        @return  The number of times anEntry appears in the bag. */
   public int getFrequencyOf(T anEntry);

   /** Tests whether this bag contains a given entry.
        @param anEntry  The entry to find.
        @return  True if the bag contains anEntry, or false if not. */
   public boolean contains(T anEntry);
```

Status

Compiler

```
javac "LinkedBagDemo1.java" (in directory: E:\BlackUSB\GavilanCollege\Spring\CSIS27\WeeklyExamples\srcSol\Bag)
Compilation finished successfully.
```

Linked BagDemo1 Screenshot with output

LinkedBagDemo1.java - E:\BlackUSB\GavilanCollege\Spring\CSIS27\WeeklyExamples\srcSol\Bag - Geany

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Symbols | Documents

- Classes
  - LinkedBagDemo1 [6]
    - displayBag [60]
    - main [8]
    - testAdd [43]
    - testIsEmpty [23]

LinkedBag1.java | LinkedBagDemo1.java | BagInterface.java

```java
/** A test of the methods add, toArray, isEmpty, and getCurrentSize,
    as defined in the first draft of the class LinkedBag.
    @author Frank M. Carrano, Timothy M. Henry
    @version 5.0
*/
public class LinkedBagDemo1
{
    public static void main(String[] args)
    {
      System.out.println("Creating an empty bag.");
      BagInterface<String> aBag = new LinkedBag1<>();
      testIsEmpty(aBag, true);
        displayBag(aBag);

      String[] contentsOfBag = {"A", "D", "B", "A", "C", "A", "D"};
        testAdd(aBag, contentsOfBag);
        testIsEmpty(aBag, false);
    } // end main

   // Tests the method isEmpty.
   // Precondition: If the bag is empty, the parameter empty should be true;
   // otherwise, it should be false.
    private static void testIsEmpty(BagInterface<String> bag, boolean empty)
   {
      System.out.print("\nTesting isEmpty with ");
      if (empty)
         System.out.println("an empty bag:");
      else
         System.out.println("a bag that is not empty:");

      System.out.print("isEmpty finds the bag ");
      if (empty && bag.isEmpty())
          System.out.println("empty: OK.");
        else if (empty)
          System.out.println("not empty, but it is: ERROR.");
        else if (!empty && bag.isEmpty())
          System.out.println("empty, but it is not empty: ERROR.");
        else
          System.out.println("not empty: OK.");
    } // end testIsEmpty

   // Tests the method add.
   private static void testAdd(BagInterface<String> aBag, String[] content)
   {
```

C:\Windows\SYSTEM32\cmd.

```
Creating an empty bag.

Testing isEmpty with an empty bag:
isEmpty finds the bag empty: OK.
The bag contains the following string(s):

Adding the following strings to the bag: A D B A C A D
The bag contains the following string(s):
D A C A B D A

Testing isEmpty with a bag that is not empty:
isEmpty finds the bag not empty: OK.


------------------
(program exited with code: 0)

Press any key to continue . . .
```

Status | Compiler | Messages

```
javac "LinkedBagDemo1.java" (in directory: E:\BlackUSB\GavilanCollege\Spring\CSIS27\WeeklyExamples\srcSol\Bag)
Compilation finished successfully.
```